

Dietary fat and semen quality among men attending a fertility clinic

BACKGROUND The objective of this study was to examine the relation between dietary fats and semen quality parameters. **METHODS** Data from 99 men with complete dietary and semen quality data were analyzed. Fatty acid levels in sperm and seminal plasma were measured using gas chromatography in a subgroup of men ($n = 23$). Linear regression was used to determine associations while adjusting for potential confounders. **RESULTS** Men were primarily Caucasian (89%) with a mean (SD) age of 36.4 (5.3) years; 71% were overweight or obese; and 67% were never smokers. Higher total fat intake was negatively related to total sperm count and concentration. Men in the highest third of total fat intake had 43% (95% confidence interval (CI): 62–14%) lower total sperm count and 38% (95% CI: 58–10%) lower sperm concentration than men in the lowest third ($P_{\text{trend}} = 0.01$). This association was driven by intake of saturated fats. Levels of saturated fatty acids in sperm were also negatively related to sperm concentration ($r = -0.53$), but saturated fat intake was unrelated to sperm levels ($r = 0.09$). Higher intake of omega-3 polyunsaturated fats was related to a more favorable sperm morphology. Men in the highest third of omega-3 fatty acids had 1.9% (0.4–3.5%) higher normal morphology than men in the lowest third ($P_{\text{trend}} = 0.02$). **CONCLUSIONS** In this preliminary cross-sectional study, high intake of saturated fats was negatively related to sperm concentration whereas higher intake of omega-3 fats was positively related to sperm morphology. Further, studies with larger samples are now required to confirm these findings.